



Mathematics I • 1st year of Bachillerato • The real numbers • Class 3 of 6 • With Ximena

The number line with no gaps



Class video: <https://youtu.be/yQVuqjyfaM>

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. In Spain the decimal separator is a comma: the board writes 1,41 for 1.41.

Before watching the video

1. With ruler and compass. Draw a line and mark 0 and 1, 5 cm apart: that is your unit. On the stretch from 0 to 1, draw a square. Put the point of the compass on 0, open it to the other end of the diagonal and turn it until it cuts the line.

Where does it land, measured with the ruler? _____ cm, that is, _____ units.

How long should the diagonal be? Why? Can you read that number exactly on your ruler?

2. Work out without a calculator: $\sqrt{9} + \sqrt{16} = \underline{\hspace{2cm}}$ and $\sqrt{(9 + 16)} = \underline{\hspace{2cm}}$. Are they equal?

3. Your bet: with ruler and compass, will it be possible to mark any number on the line? π too? Why do you think so?

While you watch the video

1. What does it mean for a number to be irrational? What are its decimals like, and why?

2. The point the compass marks is exactly $\sqrt{2}$, even though your drawing at the start was not. Why? Write down the two reasons.

3. How would you mark $\sqrt{5}$? Draw the triangle.

4. **Your bet.** How many triangles fit in the spiral before it goes all the way round and overlaps the first?

5. Is $\sqrt{2} + \sqrt{3} = \sqrt{5}$? Explain why not, in two ways. What does it have to do with your calculation 2 from "Before watching the video"?

6. **Your prediction.** A wheel of diameter 1 rolls one turn without slipping. Which number will the mark land on? Why exactly there?

7. If the diameter of a wheel doubles, what happens to the distance round? And to the distance round divided by the diameter? Why?

8. " π equals $22/7$." What is wrong? Why does $22/7$ have the same decimals as $1/7$?

9. What cannot be proved about the line? What is it, according to Dedekind?

What the *Theaetetus* tells us. Separate it into three columns.

What the text says	What is deduced	What is a hypothesis

Classify. Tick each set that each number belongs to. Careful, there are traps.

Number	Natural	Integer	Rational	Irrational	Real
0.999...					
$\sqrt{4}$					
$22/7$					
-3					
$\sqrt{5}$					
$1 + \sqrt{2}$					
π					

The construction, with gaps

Mark $\sqrt{13}$ on the line, filling in the gaps without looking at the video.

1 • Decompose. $13 = \underline{\hspace{1cm}}^2 + \underline{\hspace{1cm}}^2$.

2 • The triangle. A right triangle with legs $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$. One on the line, from 0 to $\underline{\hspace{1cm}}$. The other, of height $\underline{\hspace{1cm}}$, standing up on $\underline{\hspace{1cm}}$. By $\underline{\hspace{1cm}}$'s theorem, the hypotenuse measures $\underline{\hspace{1cm}}$.

3 • The compass. Put its point on $\underline{\hspace{1cm}}$ and open it to $\underline{\hspace{1cm}}$. When you turn it, the arc cuts the line between $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$.

Now, in your own words:

1. Why is that point exactly $\sqrt{13}$, even though you cannot write out all its digits?

2. How many triangles of the spiral would you need to reach $\sqrt{13}$?

3. Is there another way of marking $\sqrt{13}$ with a single triangle with whole-number legs? Why?

Practice

1. Mark $\sqrt{10}$ with a compass, using a single triangle. Hint: $10 = 9 + 1$.
2. Put in order from smallest to largest: $\sqrt{2}$, 1.41, $99/70$ and 1.4142. Try it without a calculator: compare their squares.
3. Which is closer to π , 3.14 or $22/7$?
4. **One more, to revise class 1.** What fraction is 0.142857 142857...? And 3.142857 142857...?
5. **One more, to revise class 2.** Prove that $\sqrt{5}$ is not a fraction. Hint: count fives.
6. **A challenge.** With a single triangle with whole-number legs you can mark $\sqrt{2}$, $\sqrt{5}$, $\sqrt{10}$, $\sqrt{13}$ and $\sqrt{17}$. Why not $\sqrt{3}$? How would you mark it?

For those who want more: the spiral, with ruler and compass

You need a sheet of paper, a ruler, a set square, a compass and a protractor. Your unit: 2 cm.

1. **Before drawing, your prediction.** In the video, the spiral was about 9° short of a full turn (the orange wedge). In your drawing, will the same be missing? And in your classmates' drawings?
2. Draw a 2 cm segment from a point O. At its end, raise a 2 cm perpendicular and join its end to O: that is $\sqrt{2}$. At the end of each hypotenuse, another 2 cm perpendicular, and back to O again. Keep going up to hypotenuse $\sqrt{17}$. Along the way, check:

Hypotenuse	Measures in your drawing (cm)	Should measure (cm)
$\sqrt{4}$		
$\sqrt{9}$		
$\sqrt{16}$		
$\sqrt{17}$		

3. Measure with the protractor the angle between hypotenuse $\sqrt{17}$ and the first segment: ____°. Does it match your classmates'? Why don't they all come out the same?

4. With the compass, carry hypotenuses $\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}$ and $\sqrt{5}$ to a line that starts at O. Where do they land?

5. Does your drawing prove how many degrees are missing? What would you need to be sure?

Another way of measuring. In a city with a grid of streets, a taxi does not travel in a straight line: the distance between two corners is what you travel along the streets. On squared paper, mark a centre and all the corners at taxi distance 3 from it. How many are there? What figure do they form? Measured with the same taxi rule, how long is its distance round? And its diameter? What is the distance round divided by the diameter?

Keep thinking

There are infinitely many fractions and infinitely many irrational numbers. Are there as many of one as of the other? Can one infinity be bigger than another? Before answering, decide what "as many as" means when you cannot finish counting.